

Designing your Landing Zones, the Cloud Adoption way

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What is the Cloud Adoption Framework?

"The Microsoft Cloud Adoption Framework for Azure is a full lifecycle framework that enables cloud architects, IT professionals, and business decision makers to achieve their cloud adoption goals"

What is the Cloud Adoption Framework?

- Best practices
- Tools
- Guidance

Help you shape your technology, business, and people strategies to achieve the best business outcomes possible through your cloud adoption effort.



DEFINE STRATEGY



PLAN



READY



ADOPT



GOVERN



MANAGE



SECURE

Cloud Adoption Framework

READY

Ready

Before adoption can begin, you create a **landing zone** to host the workloads that you plan to build in or migrate to the cloud.

Conceptualize a designated area for placement and integration of resources.

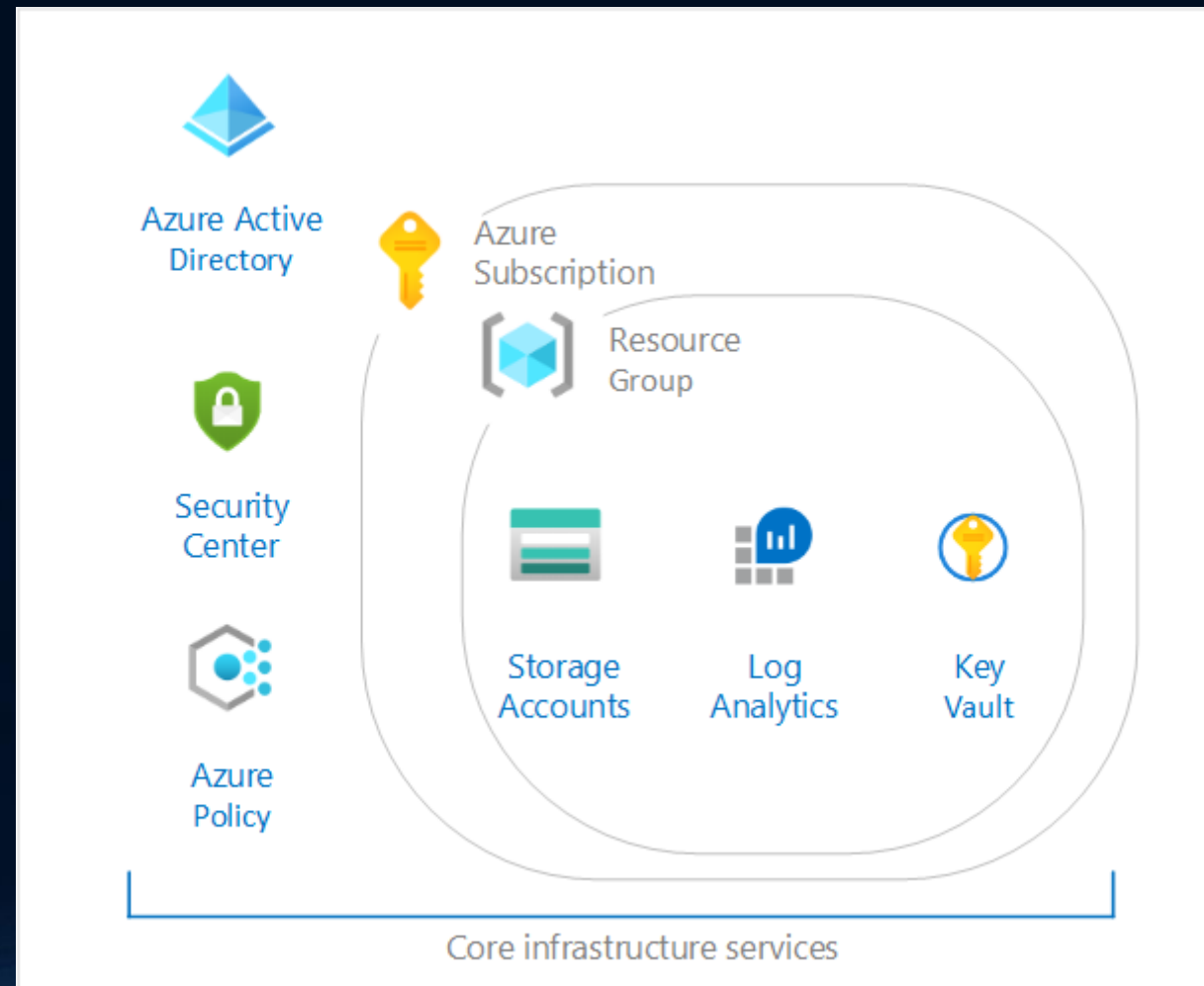
Ready

Azure landing zone implementation options

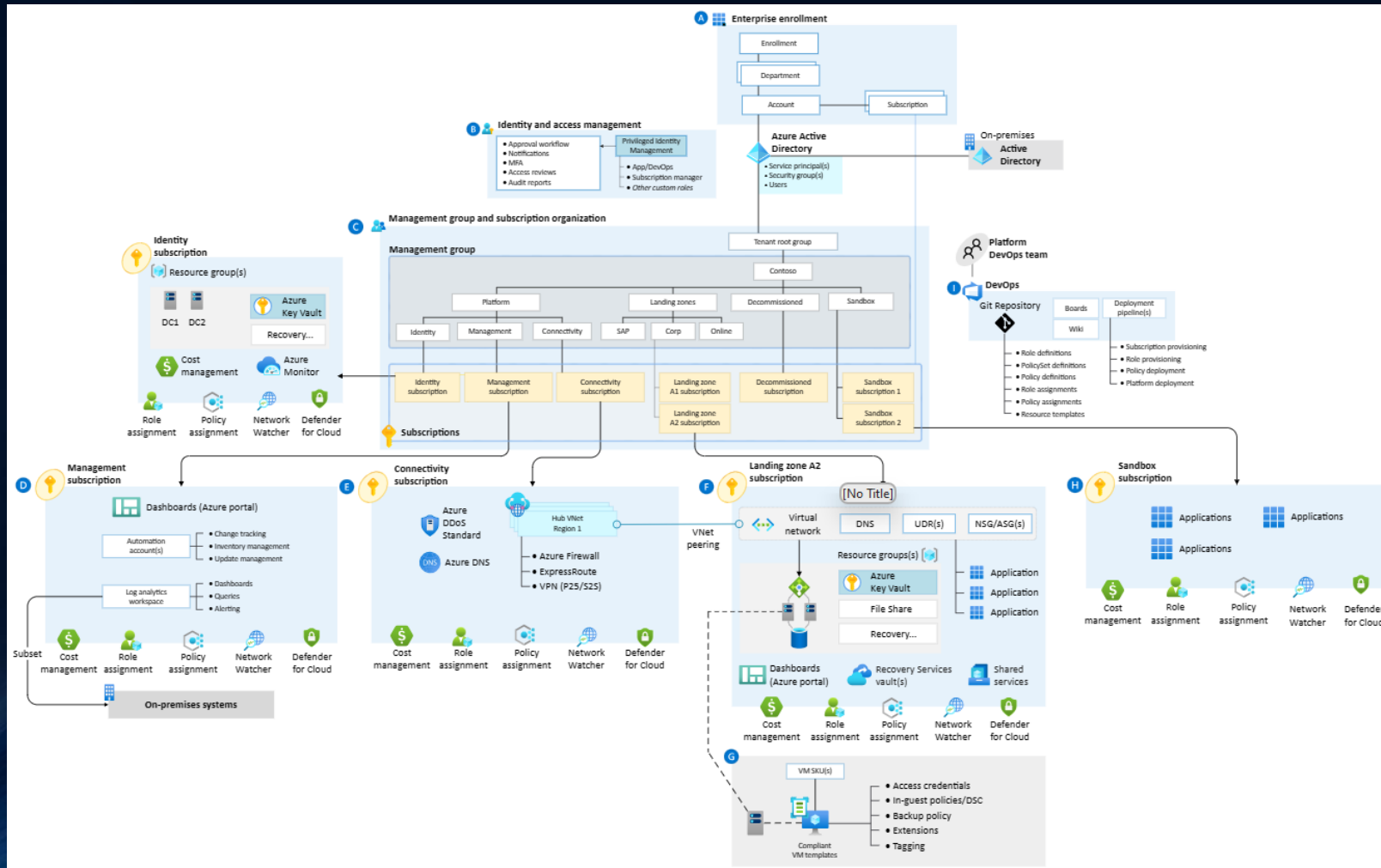
Two implementation options for landing zones:

- Start small and expand
- Enterprise-scale

Ready ALZ implementation options: Start small and expand



Ready ALZ implementation options: Enterprise-scale



Cloud Adoption Framework

READY – DESIGN AREAS

Azure billing and Active Directory tenant

Environment

Design areas

Design areas: Environment

Azure billing and Active Directory tenant

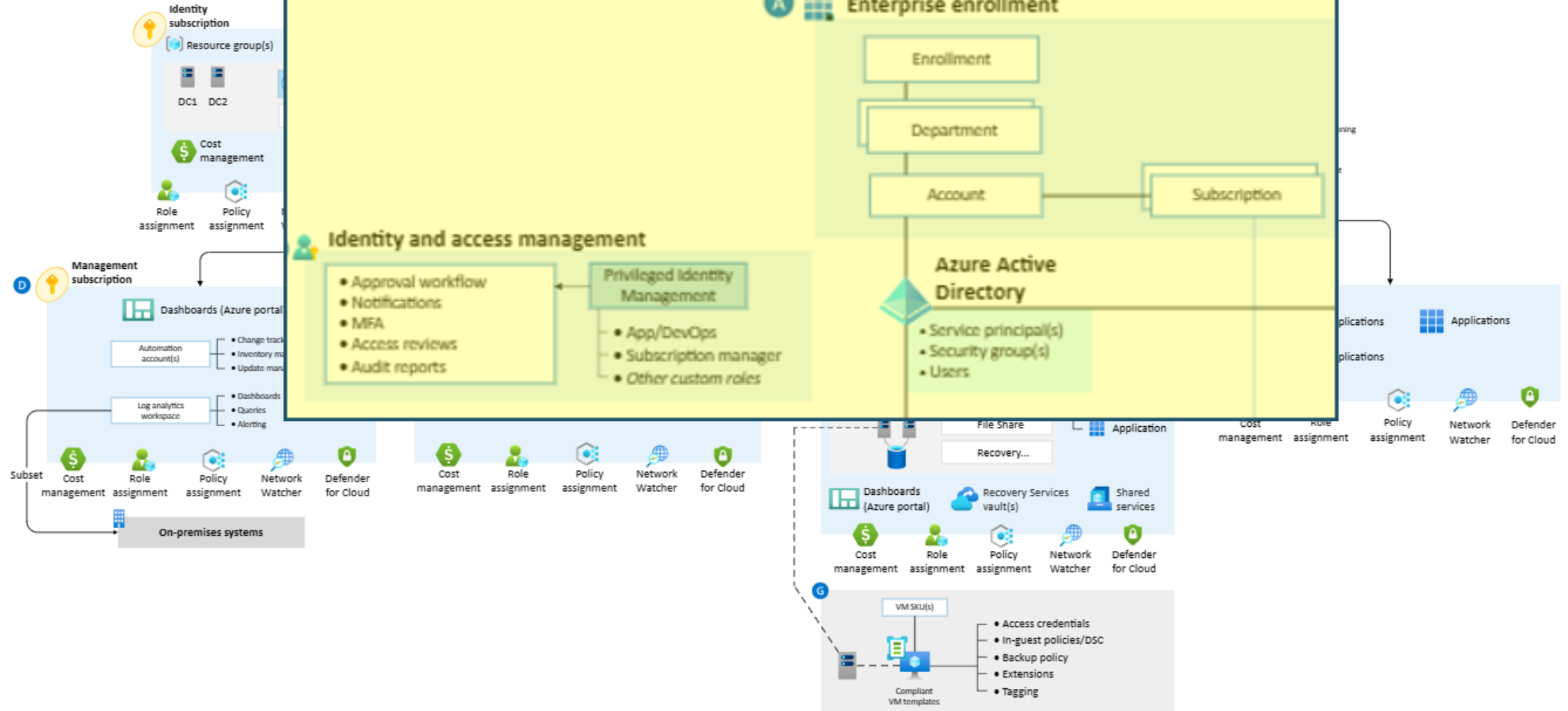
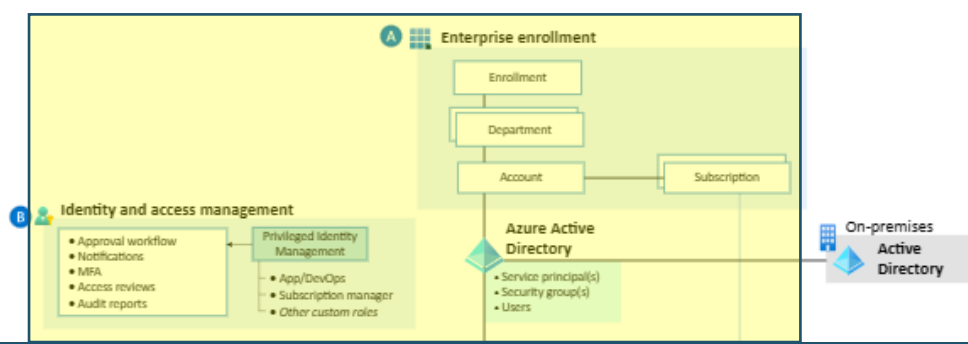
Tenant creation, enrollment, and billing setup

- Enterprise Agreement (EA)
- Microsoft customer agreement (MCA)
- Cloud solution provider agreement (CSP)

Design areas: Environment

Azure billing and Active Directory tenant

- **Custom domains**
- If no identity infrastructure, implement **Microsoft Entra-only**
- Enable **Multi Factor Authentication** (MFA)
- Enforce **conditional access policies** for privileged accounts.
- Emergency access or **break-glass accounts**
- Use **Privileged Identity Management** (PIM)
- **Diagnostic logs** to a central Azure Monitor Log Analytics workspace
- **Avoid** creating **multiple** Microsoft Entra tenants



Environment

Azure billing and Active Directory tenant

Identity and access management

Identity and access management

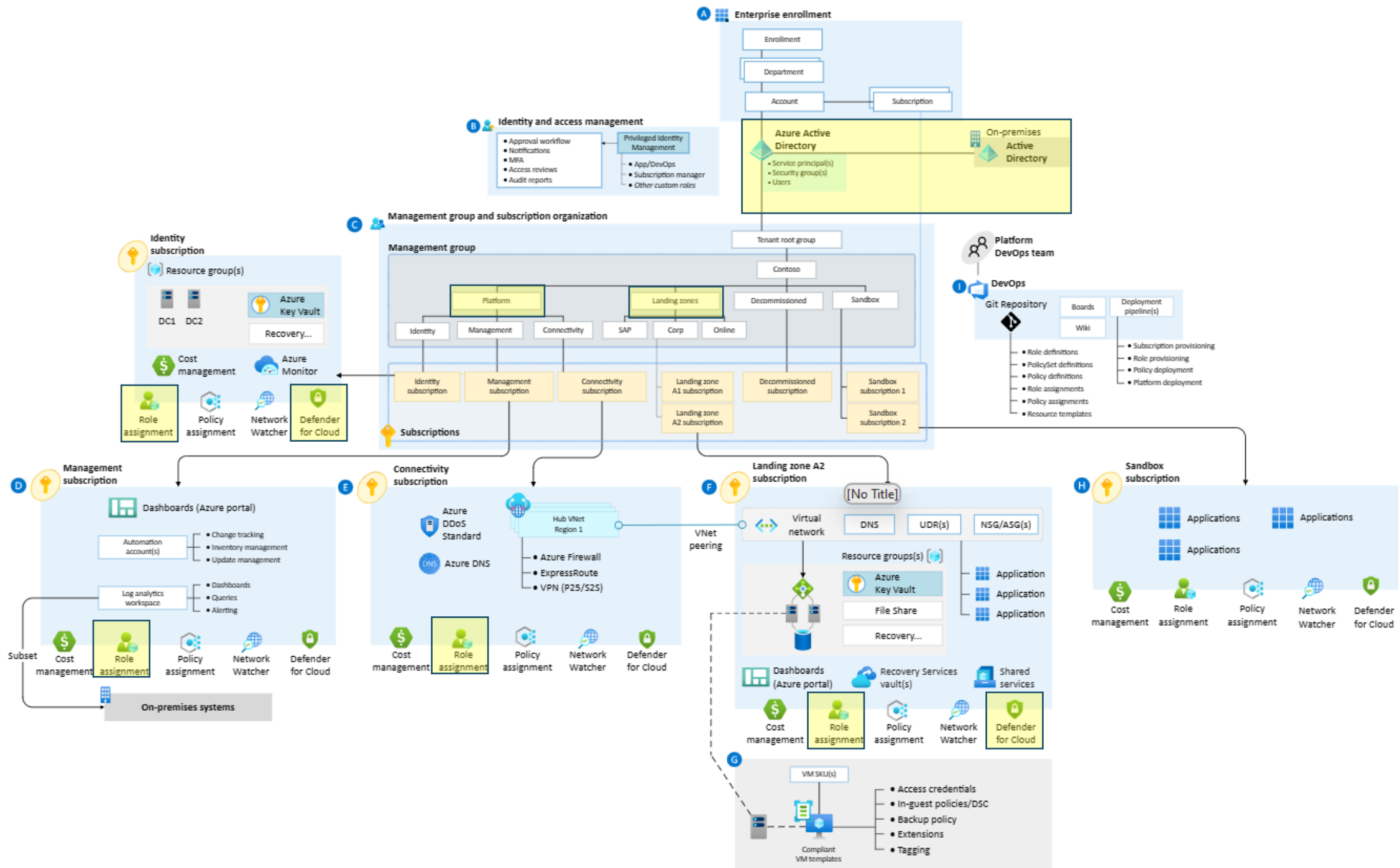
Environment

Design areas

Design areas: Environment

Identity and access management

- Microsoft Entra ID and hybrid identity
 - Integrate with on-prem AD
- Platform access
 - Grant least privilege access to platform teams
- Landing zones access
 - Grant least privilege access to application teams
 - Use Conditional Access
 - Use Managed Identities (preferred) & Service Principals
 - Use Microsoft Defender for Cloud for just-in-time access on IaaS resources



Environment

Azure billing and Active Directory tenant
Identity and access management

Resource organization

Resource organization

Environment

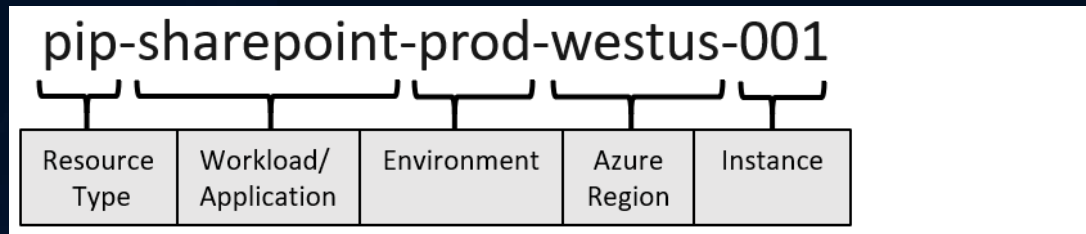
Design areas

Design areas: Environment

Resource organization: Naming & Tagging

- Define your naming convention

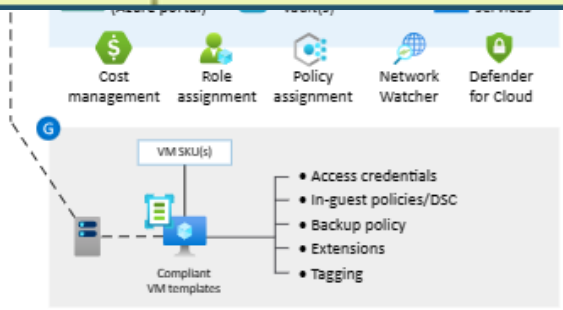
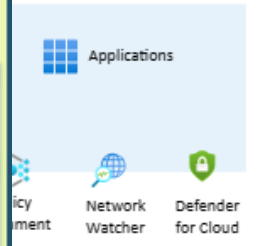
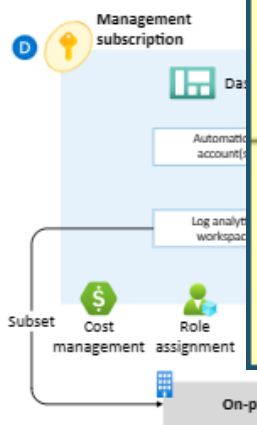
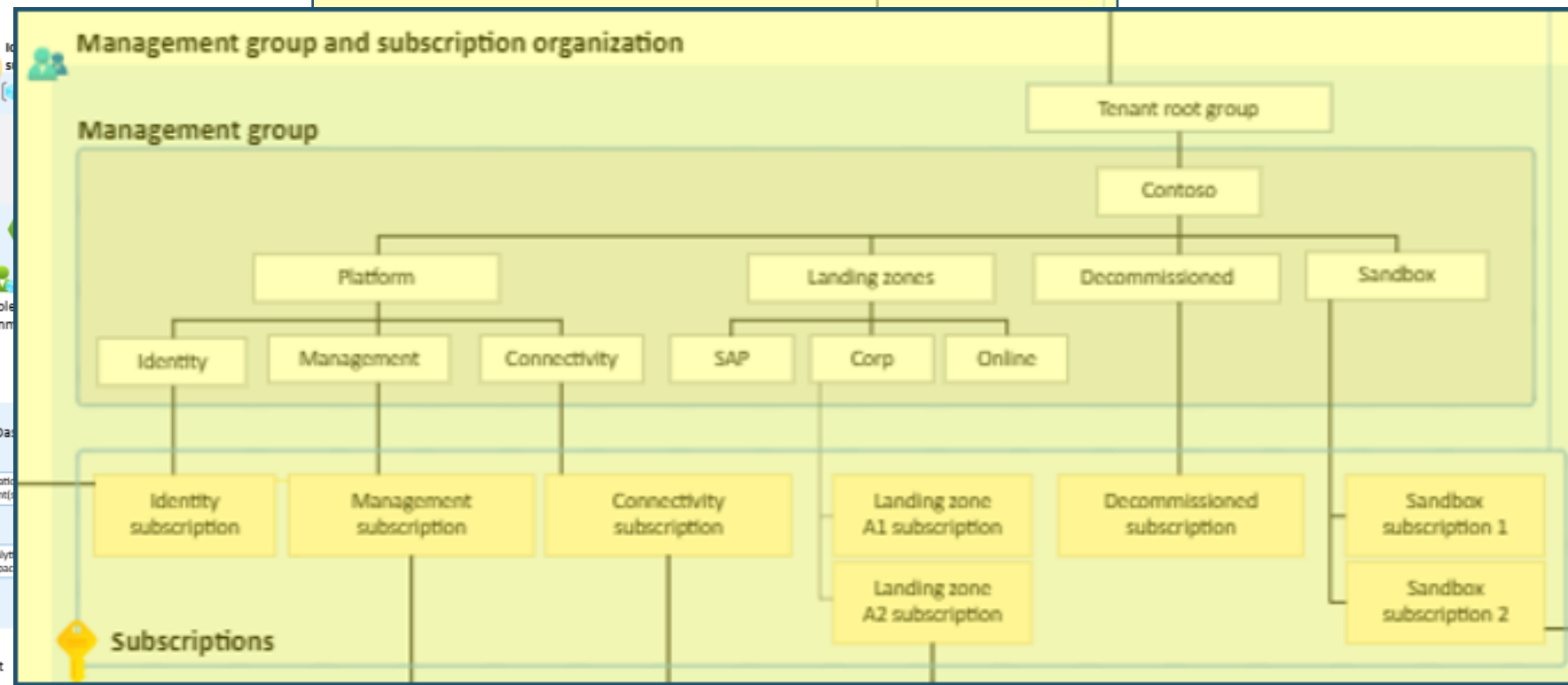
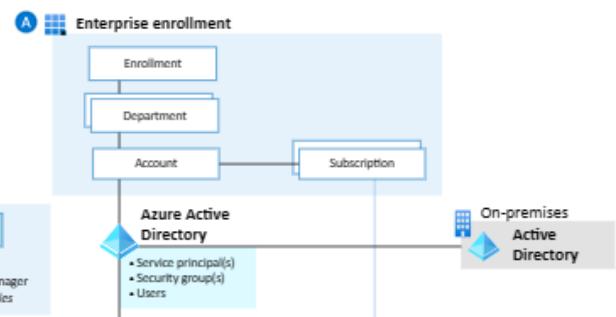
Example



- Define your tagging strategy

Example

Tag Name	Description	Key and example values
Workload name	Name of the workload the resource supports	<i>WorkloadName : ControlCharts</i>
Data classification	Sensitivity of data hosted by this resource	<i>DataClassification : Highly confidential</i>
Business criticality	Business impact of the resource or supported workload	<i>Criticality : Mission-critical</i>
Business unit	Top-level division that owns the sub scription or workload	<i>BusinessUnit : Finance</i>
Operations commitment	Level of operations support provided for workload	<i>OpsCommitment : Baseline</i>
Operations team	Team accountable for day-to-day operations	<i>OpsTeam : Central IT</i>



Design areas: Environment

Resource organization: Management Group & Subscriptions

There are two types of landing zones:

Platform landing zone

Centralized enterprise-scale foundational services for workloads and applications.

Provides shared services (identity, connectivity, management, security) to applications in application landing zones.

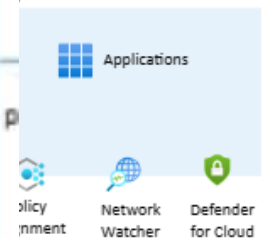
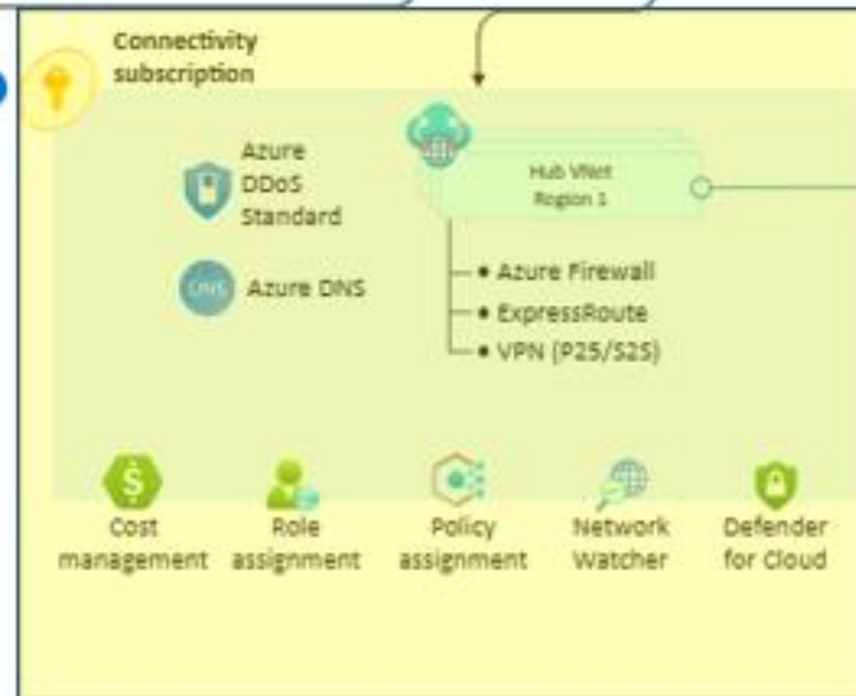
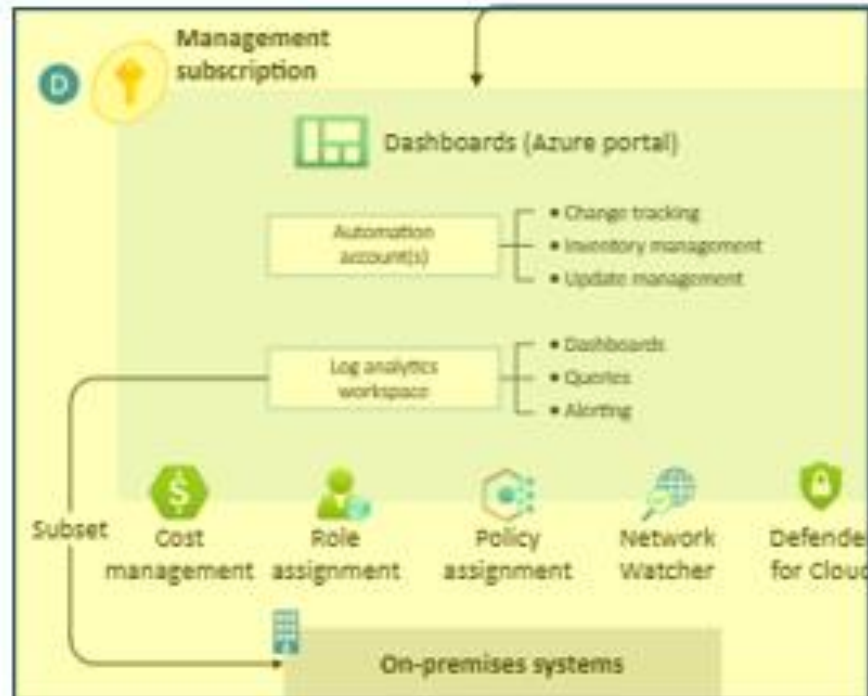
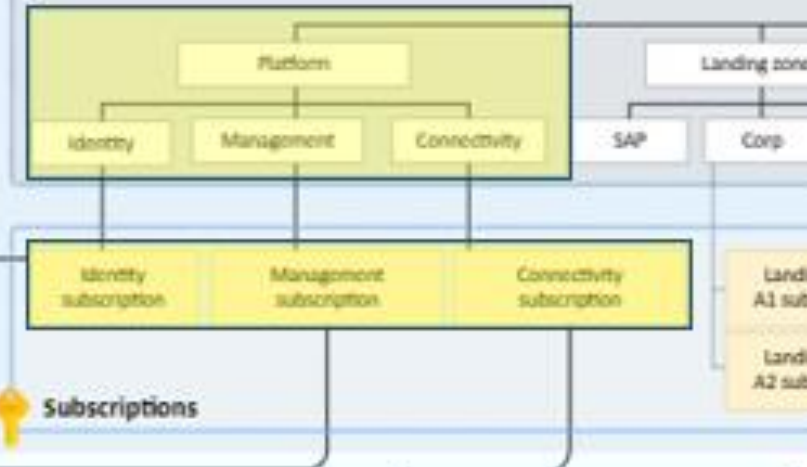
Application landing zone

Provides services specific to an application or workload.

Pre-provision application landing zones through code and use management groups to assign policy controls to them.



Management group

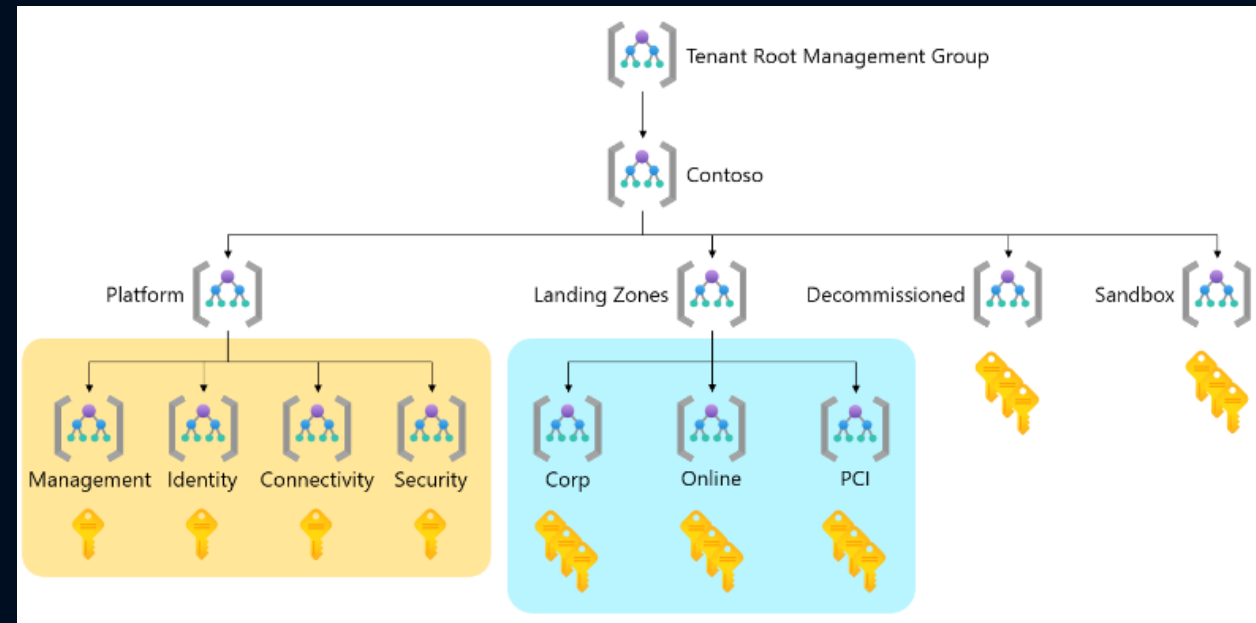


Design areas: Environment

Resource organization: Management Group & Subscriptions

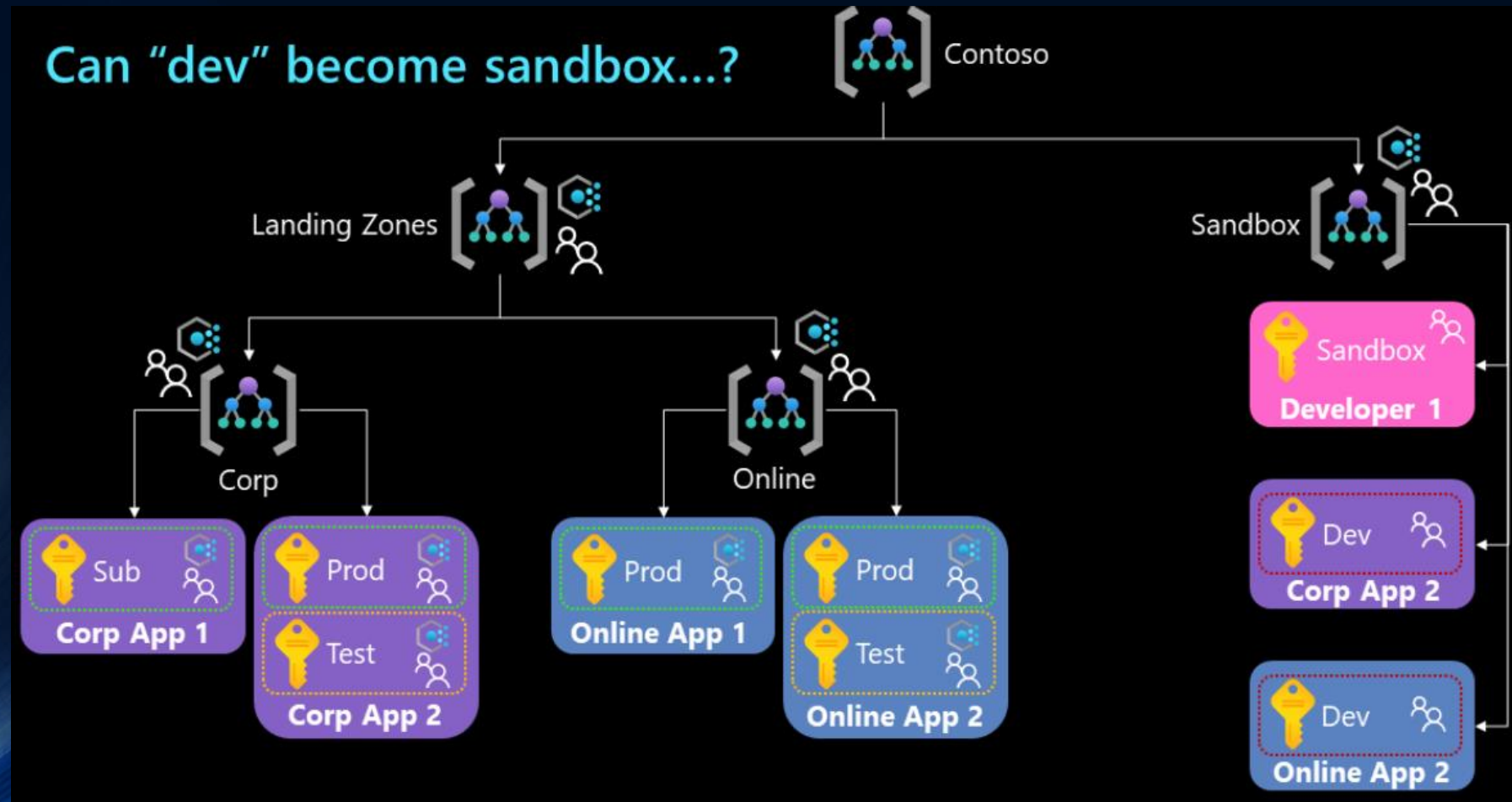
Tailoring landing zone archetypes

- Platform
 - Example: Security workload for SOC team access
- Application
 - Corp & Online
 - Some organizations need to add more
Example: PCI Workload
 - Others won't find them relevant to their organization



Design areas: Environment

Resource organization: Management Group & Subscriptions



DEMO



Environment

Azure billing and Active Directory tenant

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Resource organization

Network topology and connectivity

Network topology and connectivity

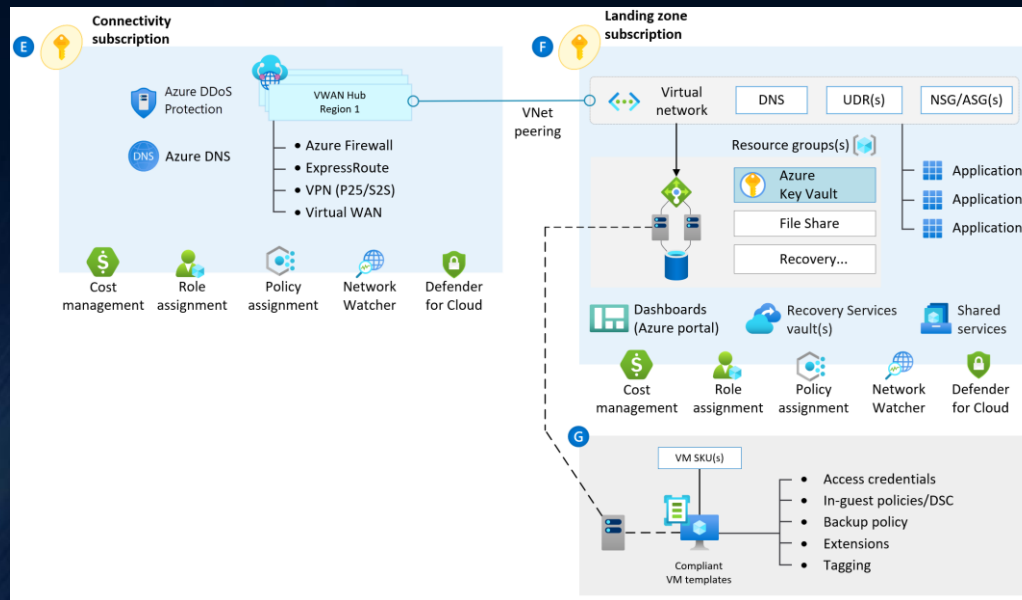
Environment

Design areas

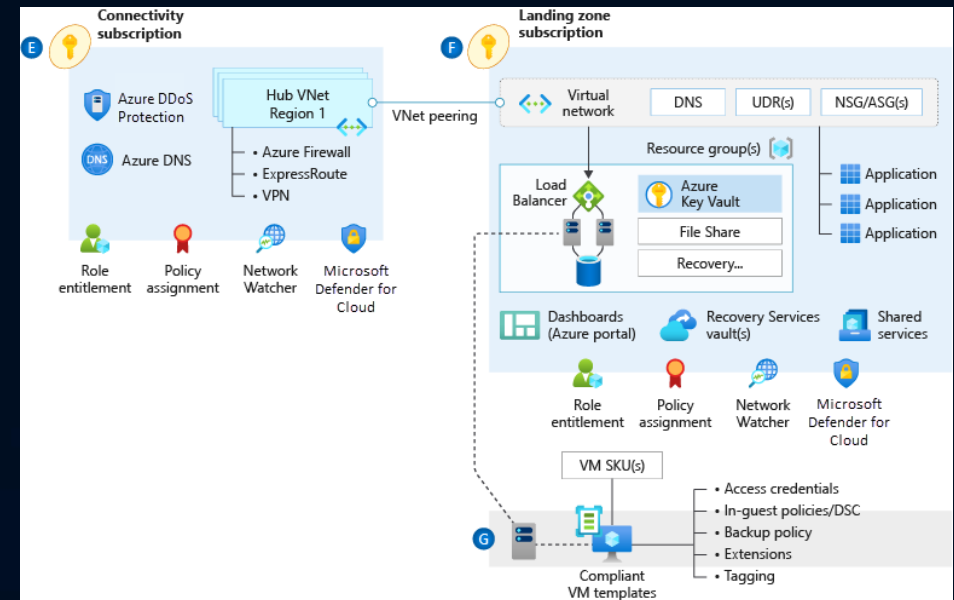
Design areas: Environment

Network topology and connectivity: Topology

- Virtual WAN network topology (Microsoft-managed)



- Traditional Azure networking topology



- Plan for IP addressing

Design areas: Environment

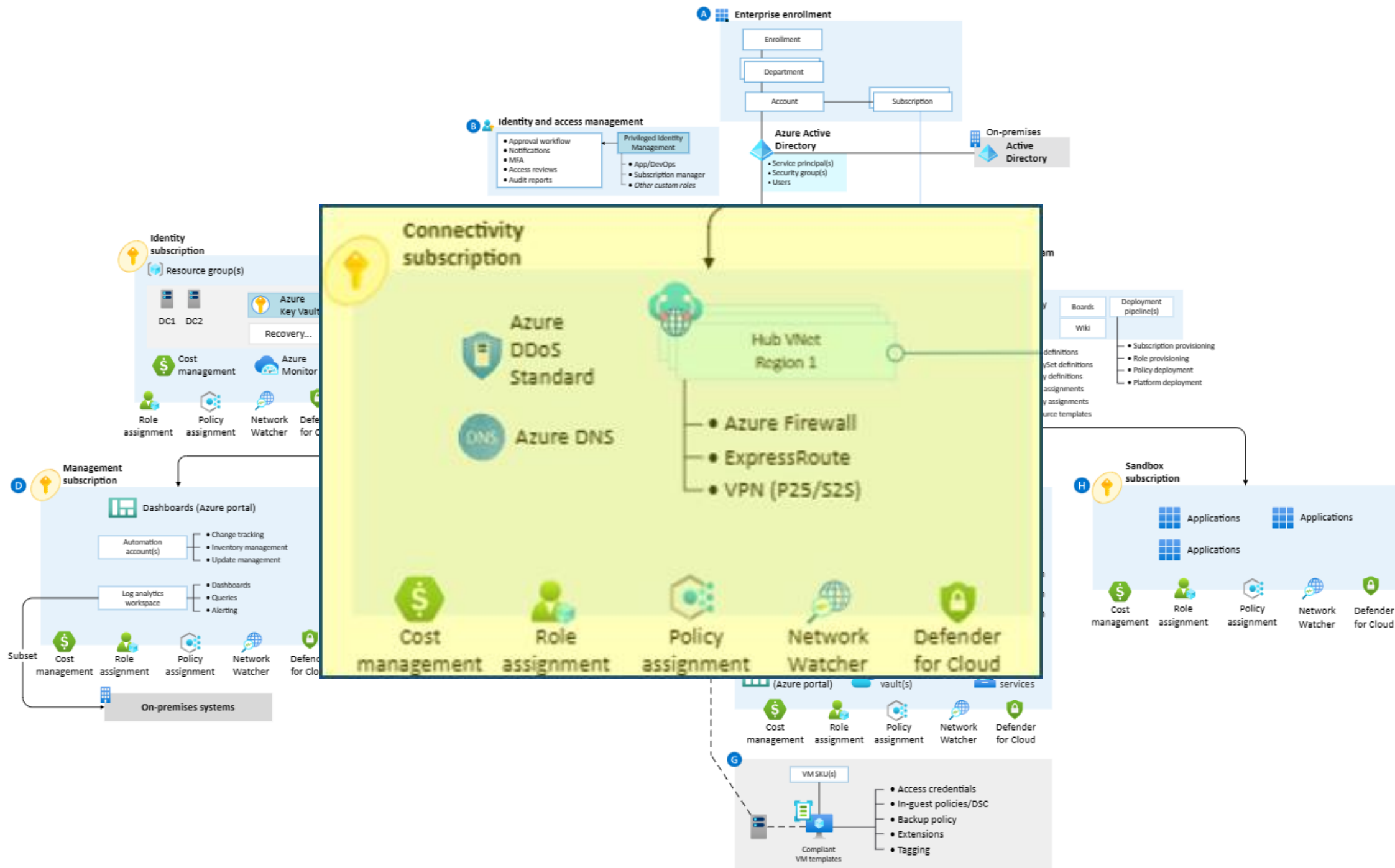
Network topology and connectivity: Connectivity

- Connectivity to Azure
 - Expressroute
 - Azure VPN (S2S)
- Connectivity to Azure PaaS services
 - Private Link
 - Private Endpoints
 - VNet Service Endpoints

Design areas: Environment

Network topology and connectivity: Network security

- Inbound and outboard internet connectivity
 - Azure Firewall
 - Azure Application Gateway Web Application Firewall
 - Frontdoor Web Application Firewall
 - NAT Gateway
- Network segmentation
 - NSG / ASG / NSG Flow logs
- Network encryption
- Traffic inspection
 - Azure Network Watcher



Security

Compliance

Design areas

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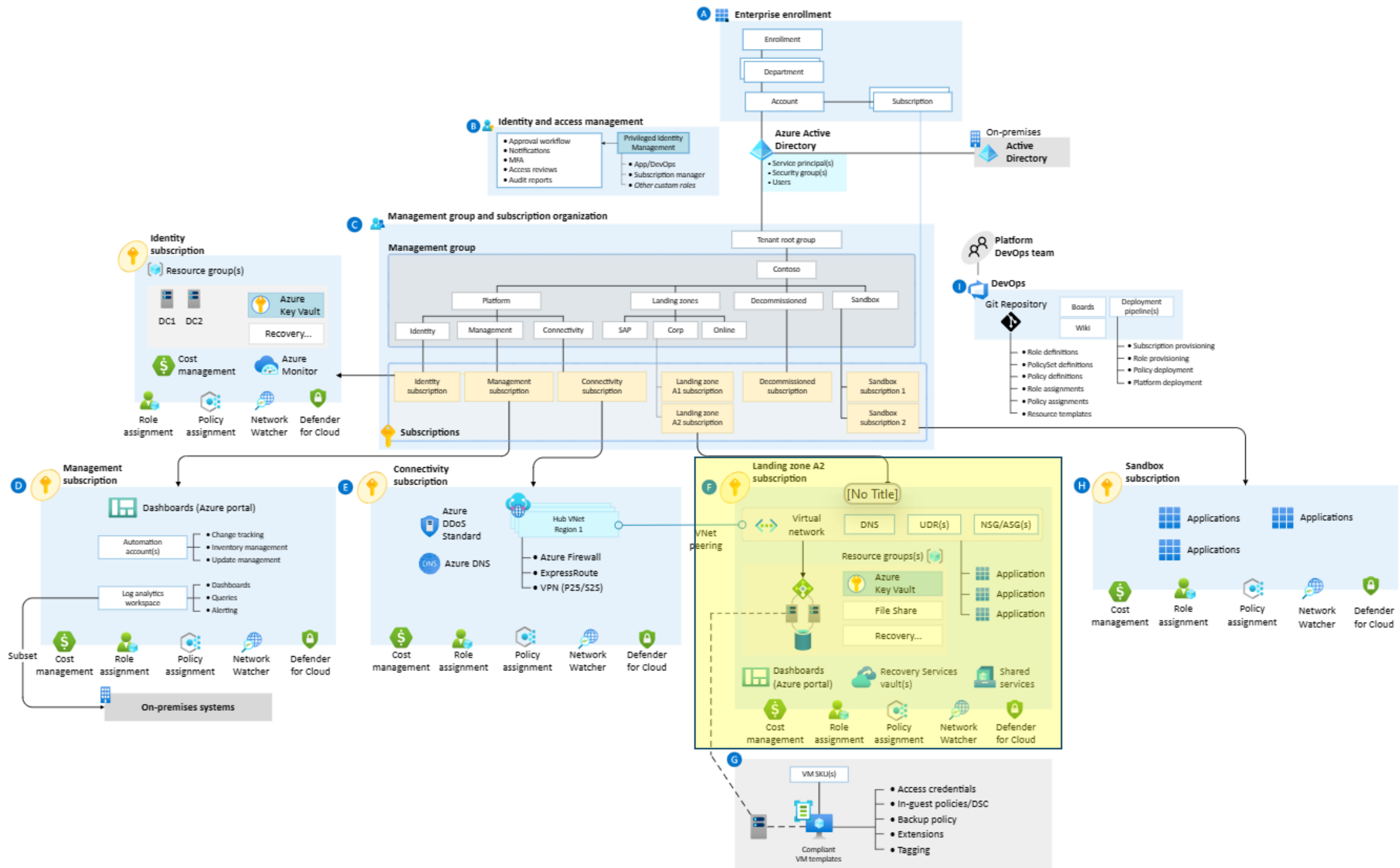
Compliance

Security

Design areas: Compliance Security

Using the **Secure methodology** of the Cloud Adoption Framework

- Design
 - Security alerts
 - Security logs
 - Security controls
 - Vulnerability management
 - Shared responsibility
 - Encryption and keys
- Operations
 - Entra ID reporting recommendations
 - Entra ID log export to Log Analytics Workspace or Storage.
 - OS patching



Management

Compliance

Design areas

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Compliance

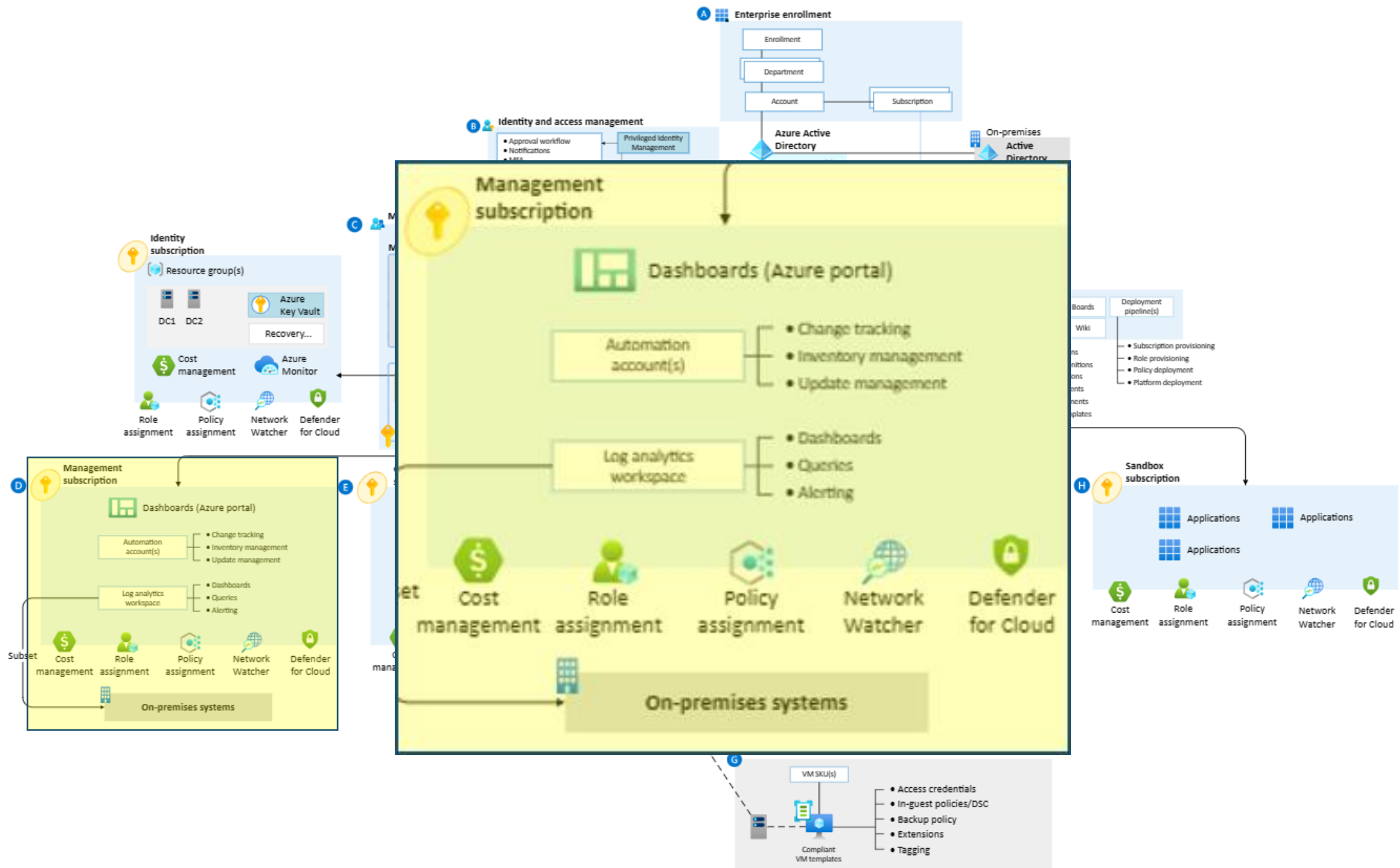
Security

Management

Design areas: Compliance Manage

Using the **Manage methodology** of the Cloud Adoption Framework

- Operations Baseline -> [Azure Monitor Baseline Alerts](#)
 - Inventory & visibility
 - Operational Compliance
 - Protect & Recover
- Advanced Operations
 - Platform Management
 - Workload Management



DEMO



Environment

Azure billing and Active Directory tenant
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Compliance

Security
Management

Governance

Governance

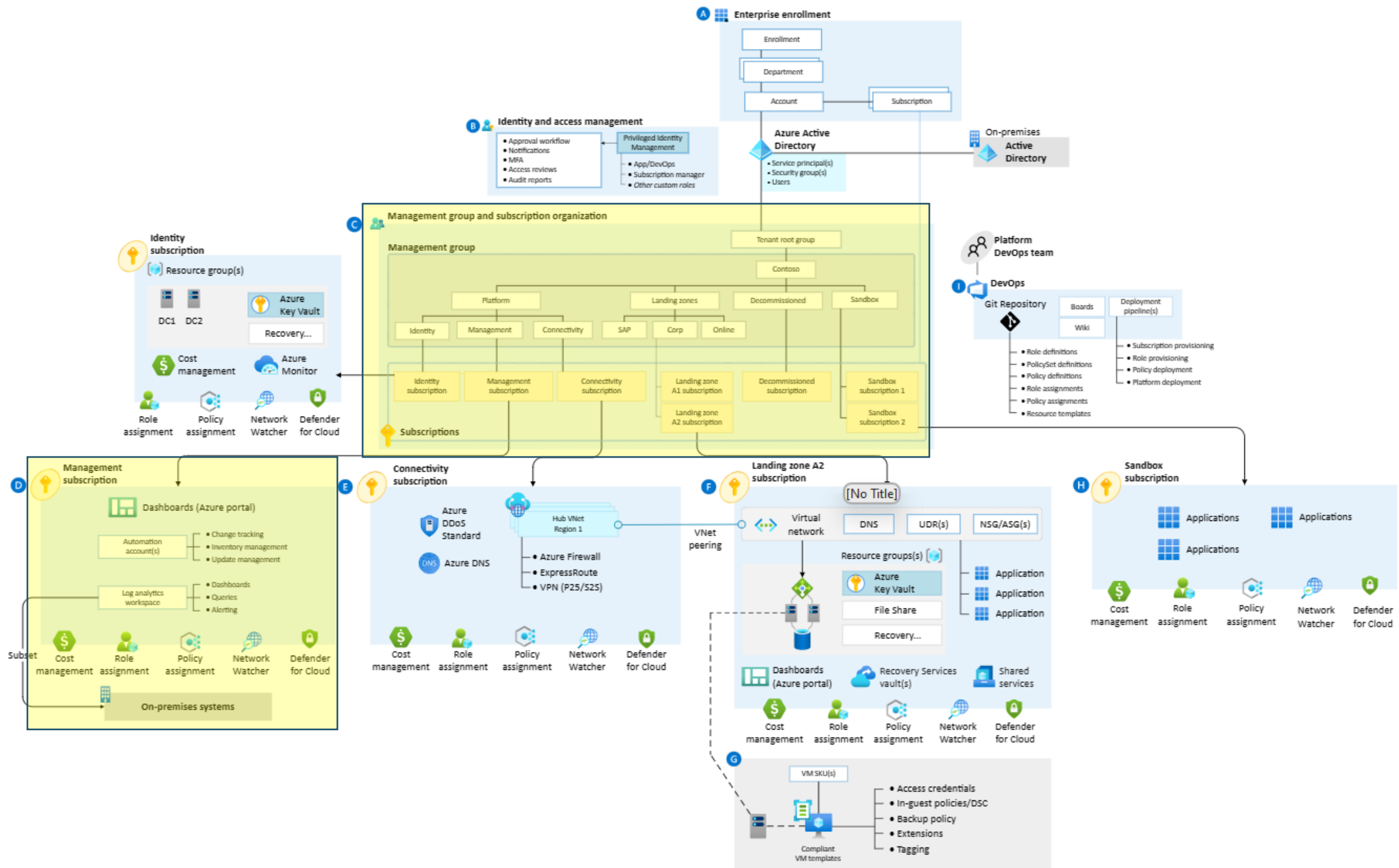
Compliance

Design areas

Design areas: Compliance
Governance

Using the Govern methodology of the Cloud Adoption Framework

- Cost management
 - Visibility / Accountability / Optimization
 - Tagging
- Resource consistency
- Identity baseline
- Deployment acceleration



Environment

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Compliance

Security
Management
Governance

Platform automation and DevOps

Platform automation and DevOps

Compliance

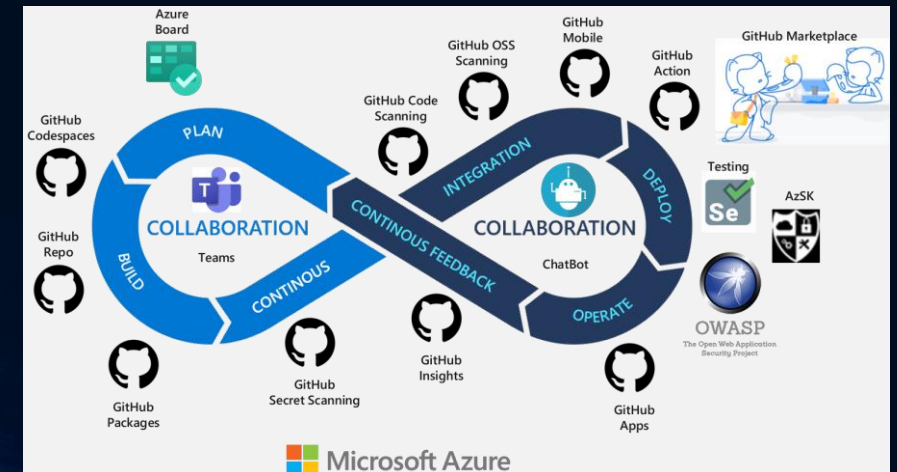
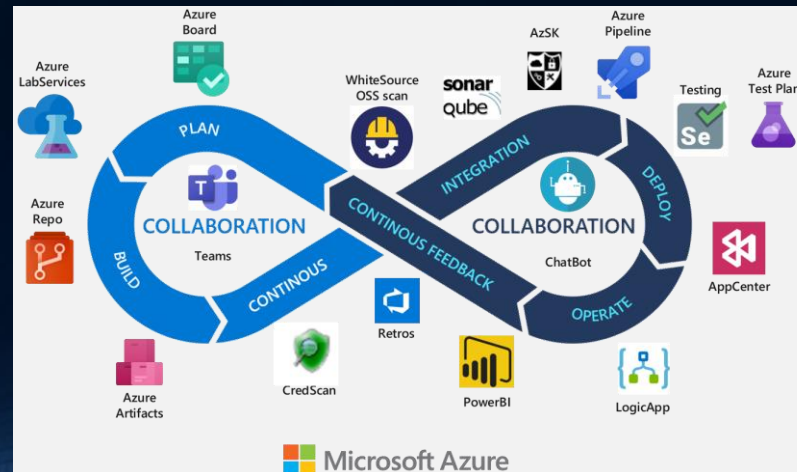
Design areas

Design areas: Compliance

Platform automation and DevOps: Automation & DevOps

- Automation
 - Everything as Code (EaC)
 - Infrastructure as Code
 - Policy as Code
 - Configuration as Code
 - Deployment as Code
 - Documentation as Code

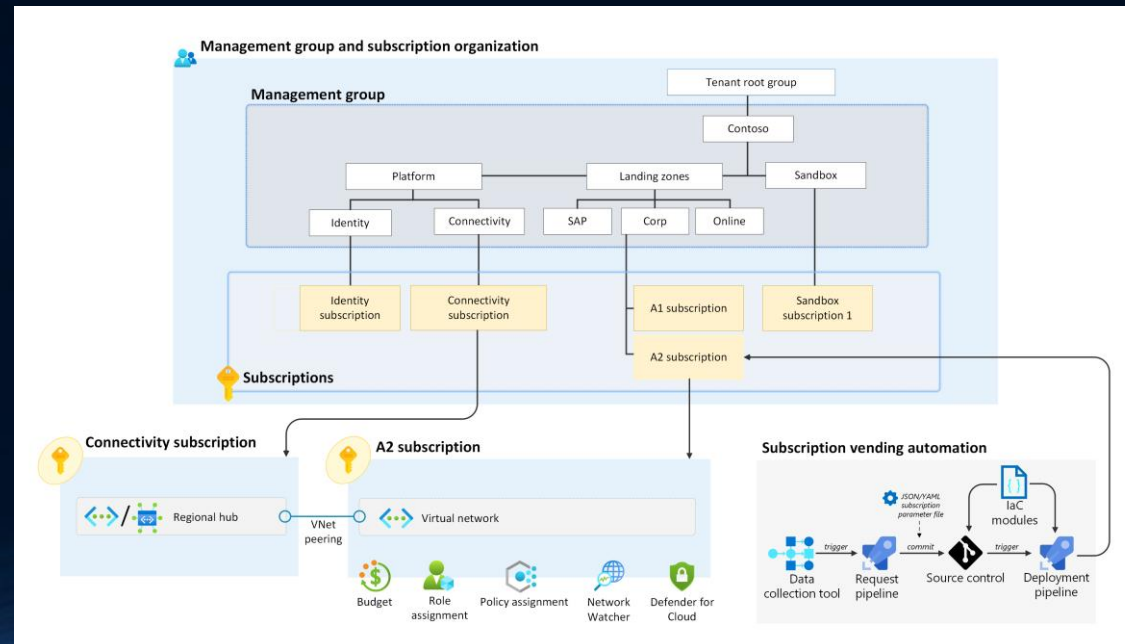
- DevOps

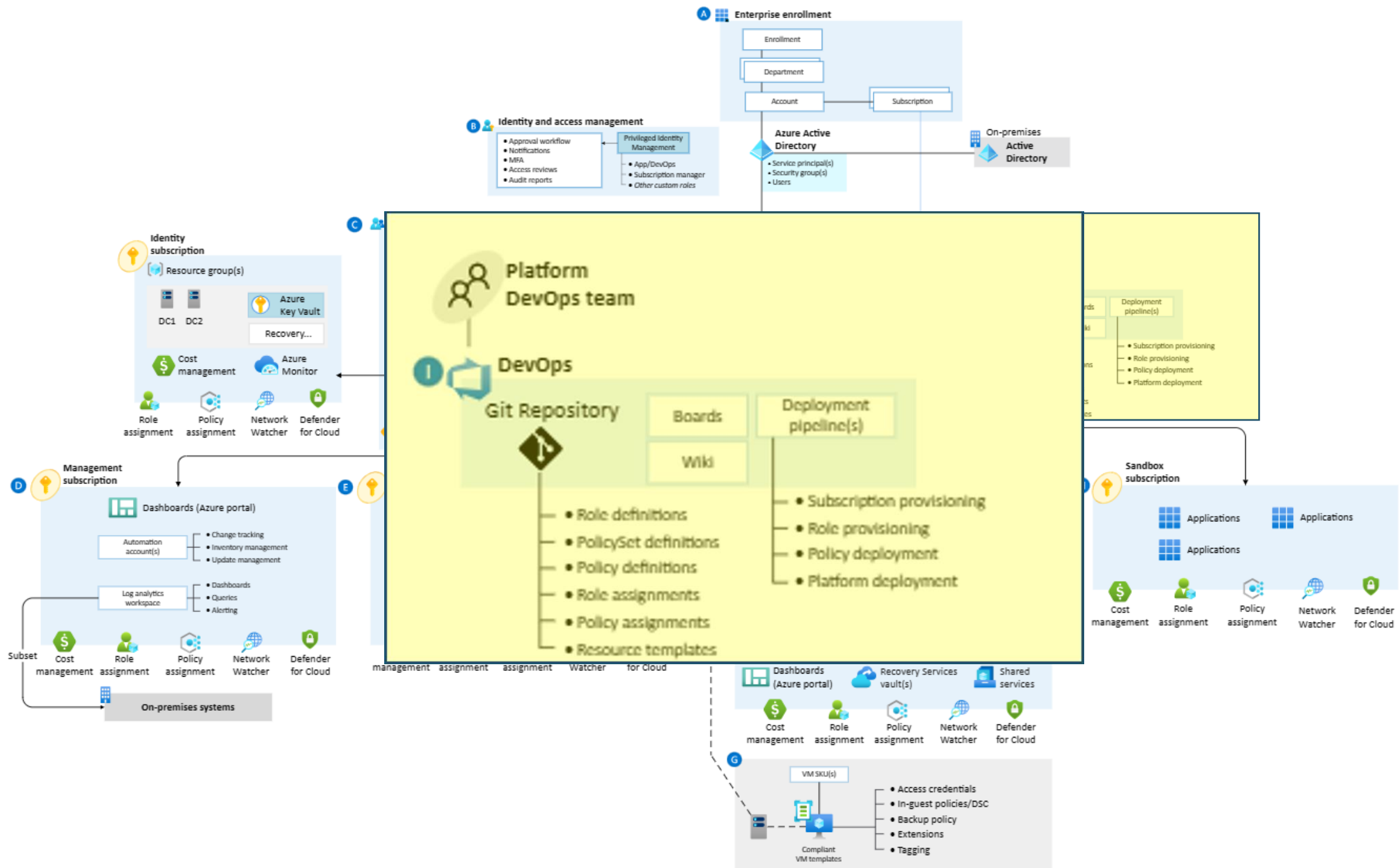


Design areas: Compliance

Platform automation and DevOps: Subscription Vending

Provides a platform mechanism for programmatically issuing subscriptions to application teams that need to deploy workloads.





Cloud Adoption Framework

READY – LANDING ZONE JOURNEY

Ready

Azure landing zone development approaches

- **Azure landing zone accelerator**

Rich initial implementation of landing zones.

Fully integrated governance, security, and operations from the start.

If needed, modify using Infrastructure-as-Code (IaC)

- **Customize**

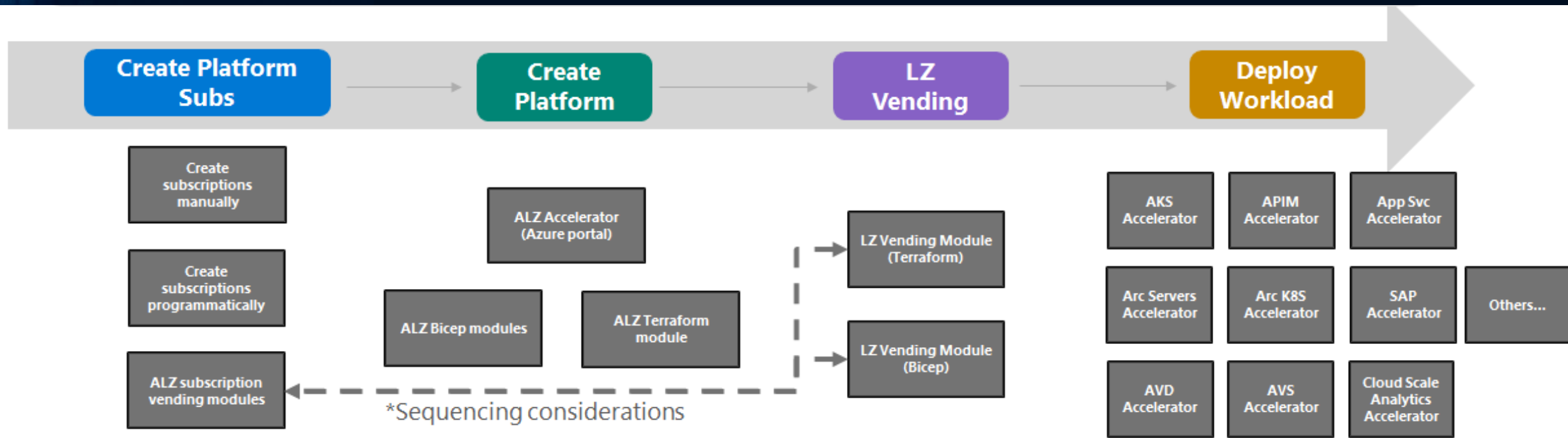
Build your environment to meet specific requirements.

Focus on the basic landing zones considerations.

Technical and business requirements are considered complete when it aligns with Azure landing zone conceptual architecture.

You can then focus on enhancing your landing zone.

Ready Landing zones journey



Cloud Adoption Framework

READY – RESOURCES

- Platform landing zone
 - [Portal](https://aka.ms/caf/ready/accelerator)
<https://aka.ms/caf/ready/accelerator>
 - [Azure landing zone Terraform accelerator](#)
 - [Azure landing zone Bicep accelerator](#)

Resources

- Application landing zones
 - [Azure Architecture center](#)

Resources

- Subscription vending
 - [Bicep](#)
 - [Terraform](#)



Thank you!